Construction of ovoid coordinate systems to describe ovoid spacetimes

Juan A. Guarín-Rojas*

Escuela de Física, Universidad Industrial de Santander, CP 680002, Bucaramanga, Colombia.

Antonio C. Gutiérrez-Piñeres†

Escuela de Física, Universidad Industrial de Santander,

CP 680002, Bucaramanga, Colombia and

Instituto de Ciencias Nucleares, Universidad Nacional Autónoma de México,

AP 70543, México, CDMX 04510, México

Rafael Torres‡

Escuela de Física, Universidad Industrial de Santander, CP 680002, Bucaramanga, Colombia

(Dated: July 27, 2026)

In this work, some Cartesian ovoid coordinates are constructed to facilitate the description of ovoid spacetimes. In optics, Cartesian ovoids are very useful for constructing optical systems without both geometric and chromatic aberrations. In astrophysics, these ovoids may be useful for describing objects with mass quadrupoles, such as merging binary stars. Then, ovoid spacetimes could be of interest to the scientific community. To construct ovoid coordinates, two special families of ovals are considered: confocal and triconfocal ovals. In the first family, the ovals share two foci, and in the second one, they share three. For both cases, another family of curves is found such that any of its members intersects the curves of its respective family of ovals at right angles. Then, some 2D coordinates are inferred from these families of curves. These coordinates are extended to 3D, yielding cylindrical and rotational-like coordinates. From the Euclidean line element in ovoid coordinates, a metric ansatz for describing a stationary ovoid spacetime is proposed. Next, this ansatz is used to obtain an exact solution of the Einstein field equations for a static anisotropic fluid.

I. INTRODUCTION

Cartesian ovoids are quartic surfaces that have been proven to be useful in optics. They were found as a solution to the problem of rigorous stigmatism [1], in which light rays emitted from a point source must be refracted to a single point by an optical surface, which turns out to be a Cartesian ovoid [2, 3]. In a plane containing the focal axis, one obtains a Cartesian oval as the intersection curve between the ovoid and the plane. In recent years, Cartesian ovoids have been recovered by Silva-Lora and Torres [4–6]. These authors use ovoids to construct singlet lenses with aplanatism and compound lenses with achromatic stigmatism, i.e., stigmatic systems free of chromatic aberration. These promising results show the importance of ovoids in optics.

Ovoids can also be considered as a specialization of the family of superconic surfaces [7–9]. These surfaces satisfy an algebraic equation of nth degree that generalizes the Schwarzschild equation for conic surfaces[1] [9, 10]. They have been proposed for use in optical design. Ovoids are also close to conic surfaces in that their respective curves, ovals and conic sections, both have foci. However, ovals have three foci [11–13]. When one of the foci goes to infinity, the Cartesian ovals become conic sections [11, 13], or a portion of them, and when the third focus coincides with the first or the second, they become limaçons of Pascal [13]. In this way, ovoids generalize conic surfaces.

In general relativity, symmetries are key to simplifying the description of spacetimes. The most common symmetries used to model astrophysical objects are spherical and axial symmetry. Models with these symmetries are widely found in the literature [14–17]. Within axially symmetric spacetimes, there are the ellipsoidal spacetimes formulated by Krasiński [18]. Some authors have found solutions to the Einstein field equations based on Krasiński's work, recovering known solutions and obtaining new ones [19–21]. Hence, spherical and ellipsoidal symmetry have been useful to construct spacetimes in general relativity.

Since spheres and ellipsoids are special cases of Cartesian ovoids, one wonders if ovoids can also be useful to construct spacetimes in general relativity. As ovoids resemble deformed

*Electronic address: juan2201870@correo.uis.edu.co
†Electronic address: acgutier@uis.edu.co
‡Electronic address: rafael.torres@saber.uis.edu.co

[1] Hereafter, conic surfaces refer to the surfaces of revolution of conic sections.

spheres, one would expect that an astrophysical object with this shape must have a mass quadrupole. This quadrupole would allow the object to emit gravitational waves in a perturbative regime [22]. Then, ovoids might be useful to describe the gravitational waves emitted by binary neutron stars during merger. As the stars coalesce, they get deformed and form some kind of funnel that could be modeled using an ovoid.

The purpose of this work is to construct some ovoid spacetimes as candidates to model the gravitational field of an ovoid-shaped object. To that end, Sections II-VI introduce Cartesian ovals and ovoid coordinates. In Section II, Cartesian ovals are treated as a three-foci curve, and their existence condition is shown. Sections IV and V present confocal and triconfocal ovals and their respective orthogonal curves. Based on the curves discussed above, Section VI formulates and characterizes ovoid coordinates. The transformation rules between ovoid and Cartesian coordinates, and the line elements in the new coordinates are obtained.

In general relativity, symmetry can be described in terms of isometry groups and geometric invariants such as Killing vectors [23]. This description applies to spherical, axial, and time-translation symmetries. To extend the notion of symmetry, Krasiński's pioneering work [18] defines ellipsoidal spacetimes in terms of a congruence of observers and the induced metric. Whether isometries or Killing vectors can characterize these spacetimes remains an open question. For now, this work focuses on proposing a metric ansatz rather than characterizing its symmetry. Based on Krasiński's ideas, Section VII defines ovoid spacetimes using the induced metric on confocal ovoids. In Section VIII, exact solutions to the Einstein field equations for a static anisotropic fluid in an ovoid spacetime are obtained.

II. CARTESIAN OVALS

A Cartesian oval is defined as the locus of points in the plane for which the sum, $cr + c'r'$, of distances to two fixed points, r and r', multiplied by real constants, c and c', is constant [2, 12, 13, 24]. For the curve to exist, c and c' can not be simultaneously zero. Without loss of generality, it can be considered $c \neq 0$, so that the equation of Cartesian ovals can be written as

$$r + ar' = u\,, \tag{1}$$

with a and u real constants. A well-known result about ovals is that there exists a set of three distinguishable bipolar equations that give rise to the same curve [11–13]. In this case,

the additional two equations to equation 1 are

$$\text{sign}(\tau(1-a^2))ar'' + \frac{u}{\tau}r = (\tau + \xi)\,, \tag{2}$$

$$\text{sign}(\tau(1-a^2))r'' - \frac{u}{\tau}r' = a\xi\,, \tag{3}$$

where the two fixed points are considered to be in the x-axis of the xy-plane, so that if the fixed point related to r is at the origin then the other one is at $(0,\tau)$, with τ a real constant, the real constant ξ is equals to $\frac{u^2-\tau^2}{(1-a^2)\tau}$, r'' is the distance to a third fixed point at $(0,\tau+\xi)$, and $\text{sign}(x) := x/|x|$ is the sign function. This third fixed point is considered as a *third focus* [12, 24], so that any oval is related to three foci. Equations 2 and 3 give rise to the same curve as equation 1 provided that ξ is not infinite and that $\tau+\xi$ is not zero, in which cases one, or both, of the equations 2 and 3 become undefined.

A less well-known result, not typically mentioned, is that the curve given by the equation 1 is not empty when

$$u \geq \min(1,a)|\tau| \quad \text{if} \quad a > -1\,, \tag{4}$$

$$u \leq |\tau| \quad \text{if} \quad a < -1\,, \tag{5}$$

$$-|\tau| \leq u \leq |\tau| \quad \text{if} \quad a = -1\,, \tag{6}$$

where $\min(1,a)$ is the minimum value between 1 and a. These are the conditions for the existence of ovals. In the case $a > -1$, these results coincide with [24]. In the appendix X, a proof of these conditions is shown. Although the existence conditions are only explicitly mentioned in [24], as far as the authors know, they are in fact implicit in the work of Descartes [2]. As ovals are classified in genres based on the procedure to draw them, a certain oval may exist only if it is possible to construct it with ruler and compass, i.e., if the required distances and radii are positive, if the intersections between lines exist, if the lines can be drawn, etc.

III. ORTHOGONAL SYSTEMS OF COORDINATES BASED ON OVALS

Consider the rectangular Cartesian coordinates x, y in the xy-plane. Any two coordinates q^1, q^2, related to the rectangular ones by the equations $x = x(q^1,q^2)$ and $y = y(q^1,q^2)$, are said to be orthogonal if their metric coefficients,

$$g_{ij} = \frac{\partial \mathbf{r}}{\partial q^i} \cdot \frac{\partial \mathbf{r}}{\partial q^j}\,, \tag{7}$$

form a diagonal matrix, where $\mathbf{r} = x(q^1, q^2)\hat{\boldsymbol{\imath}} + y(q^1, q^2)\hat{\boldsymbol{\jmath}}$ is the position vector of the point (x, y) in the xy-plane and $\hat{\boldsymbol{\imath}}, \hat{\boldsymbol{\jmath}}$ are the unitary Cartesian vectors. It is not difficult to see that the orthogonality condition for q^1, q^2 is

$$x_{,1}x_{,2} + y_{,1}y_{,2} = 0\,, \tag{8}$$

where the subscripts indicate partial differentiation, $x_{,i} := \partial x/\partial q^i$ for each $i = 1, 2$.

This condition is equivalent to say that the q-constant curves, called the coordinate curves, intersect each other at right angles, as it is shown in the appendix X. Hence, to construct a 2D system of coordinates, it is sufficient to have a pair of uniparametric families of curves $\mathscr{F}_1 = \{\gamma_r\}_{r\in\mathcal{I}}$ and $\mathscr{F}_2 = \{\sigma_s\}_{s\in\mathcal{J}}$ so that if $\gamma \in \mathscr{F}_1$ and $\sigma \in \mathscr{F}_2$ then γ and σ meet at right angles. The parameters of these two families r, s may be taken as the q^i-coordinates. Naturally, to any point in the xy-plane must pass one and only one member of each family.

If, in addition, these families $\mathscr{F}_1, \mathscr{F}_2$ can be seen as the level curves of two functions f, g, respectively, then $r = f(x, y)$ and $s = g(x, y)$ are the relations between Cartesian and rs-coordinates. Their inversion, $x = x(r, s), y = y(r, s)$, would allow to compute the metric coefficients in terms of r and s. To construct *oval* coordinates, one may consider that one, or both, of the coordinate curves are ovals. In the next sections, two special families of ovals are considered: confocal ovals and triconfocal ovals. With them, various orthogonal coordinates will be built.

IV. CONFOCAL OVALS AND SYNECDOLAS

Consider the set of ovals defined by the level curves of the function

$$f(x, y) = \sqrt{x^2 + y^2} + a\sqrt{(x - \tau)^2 + y^2}\,, \tag{9}$$

with a and τ real constants. As this ovals share the same focal distance $|\tau|$, this class of ovals can be called as *confocal ovals*. Consider some curve described by the equation $y = y(x)$ such that it intersects orthogonally to any member of the family of level curves of f. It is not difficult to show that the condition of orthogonality is that

$$\frac{dy}{dx} = \frac{f_x}{f_y}\,, \tag{10}$$

where the subscripts in f indicate partial differentiation. In its differential form, this equation takes the form $f_y dx - f_x dy = 0$, which is an inexact differential equation. However it can be reduced to an exact equation by the integration factor $w = 1/y$, so that

$(f_y/y)dx - (f_x/y)dy = 0$ is exact. Hence, it can be shown that the implicit solution to the equation 10 is

$$g(x,y) = \operatorname{arcsinh}\left(\frac{x}{y}\right) + a\operatorname{arcsinh}\left(\frac{x-\tau}{y}\right) = v\,, \tag{11}$$

with v a real constant. The set of level curves of g will be called *synecdolas*. Each of this curve crosses each confocal oval at a right angle. For $a = 1$ they become sections of hyperbolas, and for $a = -1$ they become upper or lower halves of ellipses. The term *synecdola* comes from the figure of speech "synecdoque". The intention to coin this term is inspired in the closeness of names of conics to figures of speech, such as in: circle and circumlocution, ellipse and ellipsis, hyperbola and hyperbole, and parabola and parable.

Synecdolas and confocal conform a pair of families which curves mutually intersect each other at 90 degrees. The inverse relations $x = x(u,v)$ and $y = y(u,v)$, obtained by solving the equations $f(x,y) = u$ and $g(x,y) = v$ for x and y, provide the relation between Cartesian coordinates and the new coordinates u and v. To find such inverse relations start by substituting the equations

$$x = \frac{\tau\sinh t}{\sinh t - \sinh\left(\frac{v-\tau}{a}\right)}\,, \tag{12}$$

$$y = \frac{\tau}{\sinh t - \sinh\left(\frac{v-\tau}{a}\right)}\,. \tag{13}$$

Then, the equation $g(x,y) = v$ becomes $v = v$, so that it is satisfied automatically by the substitution. The remain equation becomes

$$\frac{|\tau|\left(\cosh t + a\cosh\left(\frac{v-t}{a}\right)\right)}{\left|\sinh t - \sinh\left(\frac{v-t}{a}\right)\right|} = u\,. \tag{14}$$

The expression in absolute values is equal to $|y|$, with y taken from equation 13. Hence, by looking for inverse relations with $y = y(u,v) > 0$, equation 14 becomes

$$(a\tau - u)s^{\frac{2}{a}} + e^{\frac{v}{a}}(\tau - u)s^{\frac{1}{a}+1} + e^{\frac{v}{a}}(\tau + u)s^{\frac{1}{a}-1} + e^{\frac{2v}{a}}(a\tau + u) = 0\,, \tag{15}$$

where $s = e^t$. For $a \in \mathbb{Q}$ this equation can be rearranged into an algebraic equation of certain degree (see Appendix X). For $a = 1/3$, this equation turns out to be a cubic equation in s^2 whose positive real solution for s is

$$s = \sqrt{\alpha_+^{\frac{1}{3}} + \alpha_-^{\frac{1}{3}} - \frac{(\tau-u)e^{3v}}{\tau - 3u}}\,, \tag{16}$$

where

$$\alpha_{\pm} = \pm \frac{e^{\frac{9v}{2}}}{(\tau - 3u)^2} \sqrt{\left[(\tau + 3u)(\tau - u)^3 e^{3v} - (\tau - 3u)(\tau + u)^3\right](e^{3v} - 1)} - \frac{e^{6v}}{(\tau - 3u)^3} \left[(\tau - u)^3 e^{3v} - (\tau - 3u)(\tau^2 + 3u^2)\right] . \tag{17}$$

By substituting $t = \ln s$ with this solution of s, one obtains the desired relations $x = x(u, y)$ and $y = y(u, v)$ for the points over the x-axis in the xy-plane. Besides, from the existence conditions, it is found that u must satisfy the inequation $u \geq |\tau|/3$. One also obtains that v can take any real value. For this values of u and v, the square root of equation 16 takes real positive values, see Appendix X for a rigorous proof.

V. TRICONFOCAL OVALS

In the section II, it was mentioned that any oval is related to three special points called foci. Here we are going to use this fact to construct two families of curves that intersect each other at right angles. To see this, consider the equation that relates ξ to the other parameters of the oval

$$\xi = \frac{u^2 - \tau^2}{(1 - a^2)\tau} . \tag{18}$$

From this it is possible to solve for u, so that equation 1 can be written as

$$r + ar' = \pm\sqrt{\xi\tau(1 - a^2) + \tau^2} . \tag{19}$$

Squaring both sides and solving for a gives $a_{\pm} = \frac{-rr' \pm \tau r''}{r'^2 + \xi\tau}$. This allows to define the variables

$$\alpha = \frac{-\sqrt{x^2 + y^2}\sqrt{(x - \tau)^2 + y^2} + \tau\sqrt{(x - \tau - \xi)^2 + y^2}}{(x - \tau)^2 + y^2 + \xi\tau} , \tag{20}$$

$$\beta = -\frac{\sqrt{x^2 + y^2}\sqrt{(x - \tau)^2 + y^2} + \tau\sqrt{(x - \tau - \xi)^2 + y^2}}{(x - \tau)^2 + y^2 + \xi\tau} . \tag{21}$$

The α- and β-constant curves are the set of ovals that share the same three foci, and hence they are called as *triconfocal* ovals. It is straight forward to show that this curves mutually orthogonally intersect each other, i.e., that $\nabla\alpha \cdot \nabla\beta = 0$. The property that triconfocal ovals intersect at right angles is also reported by Williamson in [12].

To find the inverse relations, $x = x(\alpha, \beta)$ and $y = y(\alpha, \beta)$, it is convenient to write

$$\alpha + \beta = -\frac{2rr'}{\xi\tau + r'^2} , \tag{22}$$

$$\alpha\beta = \frac{-\xi\tau - \tau^2 + r^2}{\xi\tau + r'^2}\,. \tag{23}$$

From equation 22 it is found that $r = -(\alpha+\beta)(\xi\tau + r'^2)/2r'$. Replacing this in equation 23 and reorganizing one gets

$$(\beta-\alpha)^2 r'^4 + 2\left(-2\tau + \xi\left(\alpha^2+\beta^2-2\right)\right)\tau r'^2 + \xi^2\tau^2(\alpha+\beta)^2 = 0\,. \tag{24}$$

The two solutions of this equation, viewed as a quadric in $\mu \equiv r'^2$, are

$$\mu_\pm = \frac{\pm 2\sqrt{\tau^2\left(\beta^2\xi - \tau - \xi\right)\left(\alpha^2\xi - \tau - \xi\right)} + 2\tau^2 + \left(-\alpha^2 - \beta^2 + 2\right)\xi\tau}{\left(\beta-\alpha\right)^2}\,. \tag{25}$$

Since $\mu_+ - \mu_- = 4\sqrt{\tau^2\left(\beta^2\xi - \tau - \xi\right)\left(\alpha^2\xi - \tau - \xi\right)}/(\beta-\alpha)^2 > 0$, one has $\mu_+ > \mu_-$. If the intersections of α- and β-constant curves occur at the same value of r', then μ_+ must be the correct solution to ensure $r'^2 > 0$. Taking $r'^2 = \mu_\pm$ it is obtained that

$$\begin{aligned} x = &\Big[\left(\pm\alpha\beta - 1\right)\sqrt{\tau^2\left(\beta^2\xi - \tau - \xi\right)\left(\alpha^2\xi - \tau - \xi\right)} \\ &- \left[\left(1 - \alpha^2 + \alpha\beta - \beta^2\right)\tau + \xi\left(\alpha^2-1\right)\left(\beta^2-1\right)\right]\tau\Big]\frac{1}{\left(\beta-\alpha\right)^2\tau}\,, \end{aligned} \tag{26}$$

$$y^2 = -\left(\xi\tau(\alpha\beta-1) - \tau^2 \mp \sqrt{\tau^2\left(\beta^2\xi - \tau - \xi\right)\left(\alpha^2\xi - \tau - \xi\right)}\right)^2 \frac{\left(\alpha^2-1\right)\left(\beta^2-1\right)}{\left(\beta-\alpha\right)^4\tau^2}\,. \tag{27}$$

where it was considered the equations $x = \frac{r'^2 - \tau^2 - r^2}{2\tau}$ and $y^2 = r^2 - \frac{(r'^2-\tau^2-r^2)^2}{(2\tau)^2}$, derived from the definitions of r and r' in terms of x and y.

The coordinates just defined are indeed a family of coordinates that depend on τ and ξ. Some interesting cases are $\xi = 0$ and $\xi = -\tau$, with τ being any nonzero real number. For this cases, the corresponding ovals became limaçons of Pascal. This can be seen as follows. For $\xi = 0$, one has $\alpha = (-r+\tau)/r'$. By the law of cosines, $r = \sqrt{r'^2 + \tau^2 + 2r'\tau\cos\phi}$, see Fig. 1. Hence, one can find

$$r' = \frac{2\tau}{\alpha^2 - 1}(\cos\phi + \alpha)\,, \tag{28}$$

which is the equation of a limaçon in its standard form. Anagously for β, one finds $r' = \frac{2\tau}{\beta^2-1}(\cos\phi - \beta)$. Therefore, α- and β-constant curves are limaçons. For $\xi = -\tau$, one can proceed as before to find $r_\pm = \frac{2a\tau}{1-a^2}(\pm 1 - a\cos\theta)$, where the case r_+ is for $a = \alpha$ and the case r_- is for $a = \beta$. The fact that limaçons are particular cases of ovals is also mentioned in [13].

Hence, triconfocal oval coordinates can be reduced to *Pascal limaçon coordinates*. As far as the authors of this article knows, coordinates of this kind, which have limaçons as

constant curves, have not been constructed before. For $\xi = 0$, equations 26 and 27 become $x = \tau$ and $y^2 = 0$ for the second sign. Since this is inconsistent, one must take first sign instead. By doing so, one gets

$$x = \frac{\tau(\alpha^2 + \beta^2 - 2)}{(\beta - \alpha)^2}\,, \tag{29}$$

$$y^2 = -\frac{4\tau^2(\alpha^2 - 1)(\beta^2 - 1)}{(\beta - \alpha)^4}\,. \tag{30}$$

This case may be considered as the simplest case of triconfocal oval coordinates.

Finally, viewing α and β as functions of x, y, ξ, and τ one finds that

$$\beta(x, y; \xi, \tau) = \alpha(-x, -y; -\xi, -\tau)\,, \tag{31}$$

$$\alpha(\tau x, \tau y; \tau\xi, \tau) = \begin{cases} \alpha(x, y; \xi, 1)\,, & \text{if } \tau > 0; \\ \beta(x, y; \xi, 1)\,, & \text{if } \tau < 0\,. \end{cases} \tag{32}$$

By analysing $\alpha(x, y; \xi, 1)$ and $\beta(x, y; \xi, 1)$, viewed as functions of x and y, one can find the range of values that α and β can take in the inverse relations (equations 26 and 27). Let $\mathcal{D}_\alpha$ and $\mathcal{D}_\beta$ be such set of values so that $\alpha \in \mathcal{D}_\alpha$ and $\beta \in \mathcal{D}_\beta$. Then, it is found that $\mathcal{D}_\alpha$ is $[-1, 1]$, if $\tau > 0$ and $\frac{\xi}{\tau} > -1$; $[-1, f\left(\frac{\xi}{\tau}\right)]$, if $\tau > 0$ and $\frac{\xi}{\tau} \leq -1$; $(-\infty, -1] \cup [1, \infty)$, if $\tau < 0$ and $\frac{\xi}{\tau} < 0$; $(-\infty, -1]$ if $\tau < 0$ and $\frac{\xi}{\tau} = 0$; and $[-f\left(\frac{\xi}{\tau}\right), -1]$, if $\tau > 0$ and $\frac{\xi}{\tau} > 0$, where

$$f(\xi) = \frac{(\xi + 1)\left(\sqrt{\xi(1 + \xi)} - \xi\right)}{\xi\left(\xi + 1 - \sqrt{\xi(1 + \xi)}\right)}\,, \tag{33}$$

is the value of $\alpha(x, y; \xi, 1)$ at $x = x_0$ and $y = 0$, with x_0 being so that $\frac{\partial\alpha(x,0)}{\partial x}\Big|_{x=x_0} = 0$ and $x_0 > 0$. Expressing the specific set $\mathcal{D}_\alpha$ for certain ξ and τ as $\mathcal{D}_\alpha(\xi, \tau)$, one has $\mathcal{D}_\beta(\xi, \tau) = \mathcal{D}_\alpha(-\xi, -\tau)$.

VI. CHARACTERIZATION OF THE VARIOUS ORTHOGONAL COORDINATES

Confocal and triconfocal ovals induced certain systems of coordinates in which the coordinate curves are one of these class of ovals. In this section we will characterize these coordinates in terms of their metric coefficients and their basis vectors. From the methods mentioned in [25], 3D coordinates will be constructed from the 2D coordinates by "rotating" or "translating" the coordinates. This process results in cylindrical- and rotational-like coordinates, respectively.

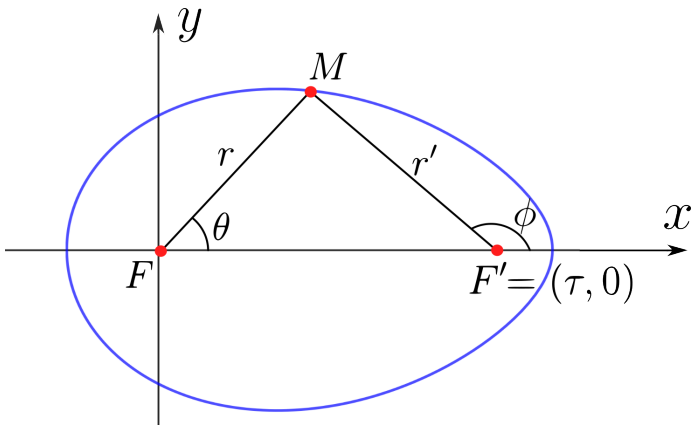


FIG. 1: A Cartesian oval shown over an xy-plane centered at the focus F. The equation of the curve is $r + ar' = u$, with $a = 0.7$ and $u = 1.2$. The focal distance is $\tau = 1$.

Consider confocal ovals as u-constant curves and synecdolas as v-constant ones. The quantities u and v define a set of *confocal oval 2D coordinates*, or simply *oval coordinates*. The metric coefficients are described in terms of the functions $x(u,v) \equiv \Lambda(u,v)$ and $y(u,v) \equiv \Upsilon(u,v)$ as

$$ds^2 = \left(\Lambda_u^2 + \Upsilon_u^2\right)\left(du^2 + \Upsilon^2 dv^2\right), \tag{34}$$

where the subscripts indicates partial differentiation. These function satisfy the relations

$$\Lambda_u = -\frac{\sqrt{\Lambda^2+\Upsilon^2}\,a\,(\tau-\Lambda) - \Lambda\sqrt{(\Lambda-\tau)^2+\Upsilon^2}}{2a\left(\Lambda^2+\Upsilon^2-\Lambda\tau\right) + \left(a^2+1\right)\sqrt{\Lambda^2+\Upsilon^2}\sqrt{(\Lambda-\tau)^2+\Upsilon^2}}\,, \tag{35}$$

$$\Upsilon_u = \frac{\Upsilon\left(\sqrt{\Lambda^2+\Upsilon^2}\,a + \sqrt{(\Lambda-\tau)^2+\Upsilon^2}\right)}{2a\left(\Lambda^2+\Upsilon^2-\Lambda\tau\right) + \left(a^2+1\right)\sqrt{\Lambda^2+\Upsilon^2}\sqrt{(\Lambda-\tau)^2+\Upsilon^2}}\,, \tag{36}$$

$$\Lambda_v = \Upsilon\Upsilon_u\,, \tag{37}$$

$$\Upsilon_v = -\Upsilon\Lambda_u\,, \tag{38}$$

which can be obtained by taking partial differentiation to the equations $u = \sqrt{x^2+y^2} + a\sqrt{(x-\tau)^2+y^2}$ and $v = \operatorname{arcsinh}(x/y) + a\operatorname{arcsinh}((x-\tau)/y)$, which define u and v in terms of x and y. In addition, the basis vectors related to u and v are

$$\hat{\boldsymbol{e}}_u = \frac{1}{\sqrt{\Lambda_u^2+\Upsilon_u^2}}\left(\Lambda_u\hat{\boldsymbol{e}}_x + \Upsilon_u\hat{\boldsymbol{e}}_y\right), \tag{39}$$

$$\hat{\boldsymbol{e}}_v = \frac{1}{\sqrt{\Lambda_u^2+\Upsilon_u^2}}\left(\Upsilon_u\hat{\boldsymbol{e}}_x - \Lambda_u\hat{\boldsymbol{e}}_y\right). \tag{40}$$

Another system of coordinates may be derived from triconfocal ovals. The quantities α and β, defined in the previous section, are consider as a set of *triconfocal oval 2D coordinates*, or *triconfocal coordinates* for short. In this case, the metric coefficients are given by

$$ds^2 = g_{\alpha\alpha}\left(d\alpha^2 + \frac{(\alpha^2-1)(\alpha^2\xi-\tau-\xi)}{(1-\beta^2)(\beta^2\xi-\tau-\xi)}d\beta^2\right), \tag{41}$$

with

$$g_{\alpha\alpha} = \Big[2\left(-\alpha\beta\xi+\tau+\xi\right)\sqrt{\tau^2\left(\beta^2\xi-\tau-\xi\right)\left(\alpha^2\xi-\tau-\xi\right)} + \big(2\left(\alpha^2\beta^2+1\right)\xi - \left(\beta+\alpha\right)^2\left(\xi+\tau\right) - 4\tau\big)\xi\tau + 2\tau^3\Big]\frac{\beta+\alpha}{\left(\alpha^2-1\right)\left(\beta-\alpha\right)^3\left(\alpha^2\xi-\tau-\xi\right)}. \tag{42}$$

The basis vectors are ... In the case $\xi = 0$, this coordinates get reduced to Pascal limaçon coordinates, whose metric coefficients are given by

$$ds^2 = \frac{4(\alpha+\beta)\tau^2}{(\alpha-\beta)^3}\left(\frac{1}{\alpha^2-1}d\alpha^2 + \frac{1}{1-\beta^2}d\beta^2\right). \tag{43}$$

Based on the two set of coordinates described before, one can easily obtain cylindrical-like coordinates. For this one just needs to introduce a z coordinate in the orthogonal direction to the xy-plane in which the 2D coordinate curves live. Hence, one obtains *confocal oval-cylinder coordinates* u, v and z, and *triconfocal oval-cylinder coordinates* α, β and z. Their line elements are respectively

$$ds^2 = \left(\Lambda_u^2 + \Upsilon_u^2\right)\left(du^2 + \Upsilon^2 dv^2\right) + dz^2, \tag{44}$$

$$ds^2 = g_{\alpha\alpha}\left(d\alpha^2 + \frac{(\alpha^2-1)(\alpha^2\xi-\tau-\xi)}{(1-\beta^2)(\beta^2\xi-\tau-\xi)}d\beta^2\right) + dz^2. \tag{45}$$

It should be pointed out that the Laplace equation admits a separable solution for the triconfocal oval-cylinder coordinates, but not for the confocal ones.

Next, rotational coordinates can be obtained in the next way. Consider a point $(x(q^1,q^2), y(q^1,q^2), 0)$ in some xyz-space related to coordinates q^1 and q^2. Rotate this point around either the x- or y-axis in a certain angle ϕ. The set of points obtained by fixing one of the quantities q^1, q^2 and ϕ define a certain surface. The resulting classes of surfaces orthogonally intersect each other and define the coordinate surfaces of certain system of coordinates. Naturally, it only make sense to rotate the points around the symmetry axis of the coordinate 2D curves. For ovals, this coincides with the focal axis.

Applying this to confocal ovals, one obtains the coordinates

$$\tilde{u} = \sqrt{x^2+y^2+z^2} + a\sqrt{x^2+y^2+(z-\tau)^2}\,, \tag{46}$$

$$\tilde{v} = \operatorname{arcsinh}\left(\frac{z}{\sqrt{x^2+y^2}}\right) + a\operatorname{arcsinh}\left(\frac{z-\tau}{\sqrt{x^2+y^2}}\right)\,, \tag{47}$$

$$\phi = \operatorname{atan2}(y,x)\,, \tag{48}$$

with inverse relations $x = \Upsilon(\tilde{u},\tilde{v})\cos\phi$, $y = \Upsilon(\tilde{u},\tilde{v})\sin\phi$ and $z = \Lambda(\tilde{u},\tilde{v})$. The corresponding line element is

$$ds^2 = \left(\Lambda_{\tilde{u}}^2 + \Upsilon_{\tilde{u}}^2\right)\left(d\tilde{u}^2 + \Upsilon^2 d\tilde{v}^2\right) + \Upsilon^2 d\phi^2\,. \tag{49}$$

One can check that $\tilde{u}-$ and $\tilde{v}-$constant surfaces are surfaces of revolution of their respective coordinate curves around the symmetry axis. Since the surface of revolution of ovals are known as ovoids, this coordinates may be called as *confocal ovoid 3D coordinates*, or simply as *ovoid coordinates*. In the same spirit, the term *synecdoloids* is coined to refer to the $\tilde{v}$-constant surfaces, in analogy to the one used for the surface of revolution of hyperbolas. The line element 49 is the one that will be use in the next section to construct stationary spacetimes with ovoid symmetry.

Finally, for triconfocal oval coordinates one obtains

$$\tilde{\alpha} = \frac{-\sqrt{z^2+x^2+y^2}\sqrt{(z-\tau)^2+x^2+y^2} + \tau\sqrt{(z-\tau-\xi)^2+x^2+y^2}}{(z-\tau)^2+x^2+y^2+\xi\tau}\,, \tag{50}$$

$$\tilde{\beta} = -\frac{\sqrt{z^2+x^2+y^2}\sqrt{(z-\tau)^2+x^2+y^2} + \tau\sqrt{(z-\tau-\xi)^2+x^2+y^2}}{(z-\tau)^2+x^2+y^2+\xi\tau}\,. \tag{51}$$

$$\phi = \operatorname{atan2}(y,x)\,, \tag{52}$$

with the inverse relations $x = \kappa(\tilde{\alpha},\tilde{\beta})\cos\phi$, $y = \kappa(\tilde{\alpha},\tilde{\beta})\sin\phi$ and $z = \eta(\tilde{\alpha},\tilde{\beta})$, where $\eta(\alpha,\beta)$ is taken to be the right hand side of the equation 26 and $\kappa(\alpha,\beta)$ is taken to be the square root of the right hand side of the equation 27. The quantity ϕ takes values in the interval $[0,2\pi)$, and as in the previous section $\tilde{\alpha}\in\mathcal{D}_{\tilde{\alpha}}$ and $\tilde{\beta}\in\mathcal{D}_{\tilde{\beta}}$. The line element is

$$ds^2 = g_{\tilde{\alpha}\tilde{\alpha}}\left(d\tilde{\alpha}^2 + \frac{(\tilde{\alpha}^2-1)(\tilde{\alpha}^2\xi-\tau-\xi)}{(1-\tilde{\beta}^2)(\tilde{\beta}^2\xi-\tau-\xi)}d\tilde{\beta}^2\right) + \kappa^2 d\phi^2\,. \tag{53}$$

This coordinates may be called as *triconfocal ovoid 3D coordinates*.

VII. OVOID SPACETIMES

The symmetries in a spacetime are usually characterized by geometric invariants such as Killing vectors. These vector can be associated to a group of isometries isomorphic to a certain group such as $SO(2)$ or $SO(3)$. In this sense, it is natural to define an axially symmetric spacetime or a spherically symmetric spacetime as one in which there is a subgroup G of the group of isometries[2] which is isomorphic to $SO(2)$ or $SO(3)$, respectively. However, for other spacetimes such as ellipsoidal or ovoid spacetimes, there is not such a natural definition since there is not a group related to ellipsoids or ovoids. The best one can do is to propose a metric ansatz that encapsulates some desirable features. One of such may be that, for example, there exits some congruence of observers in which the induced metric over their orthogonal hypersurfaces resembles or be similar to the metric over ellipsoids or ovoids. Such proposal is the one of Krasiński [18], which construct an stationary oblate-ellipsoidal metric ansatz. Here, a stationary ovoid metric ansatz will be built based on the same procedure.

In order to do that, consider a congruence of observers characterized by the four-velocity $w = U\partial_t + V\partial_\phi$. As the spacetime is consider to be stationary and axially symmetric, there exist two conmuting killing vector fields, which here are considered as ∂_t and ∂_ϕ. The

TABLE I: Orthogonal coordinates. $\mathcal{D}_u$ is given by the equation 73 and $\mathcal{D}_\alpha$ and $\mathcal{D}_\beta$ are described in the section V. For the Pascal limaçon coordinates it was made $\alpha \equiv \mu$ and $\beta \equiv \nu$.

System of coordinates	Coordinates	Conditions		
Confocal oval coordinates	u, v	$u \in \mathcal{D}_u$	$v \in \mathbb{R}$	
Confocal oval-cylinder coordinates	u, v, z	$u \in \mathcal{D}_u$	$v \in \mathbb{R}$	$z \in \mathbb{R}$
Confocal ovoid coordinates	$\tilde{u}, \tilde{v}, \phi$	$\tilde{u} \in \mathcal{D}_u$	$\tilde{v} \in \mathbb{R}$	$\phi \in [0, 2\pi)$
Triconfocal oval coordinates	α, β	$\alpha \in \mathcal{D}_\alpha$	$\beta \in \mathcal{D}_\beta$	
Pascal limaçon coordinates	μ, ν	$\mu \in [-1, 1]$	$\nu \in (-\infty, -1]$	
Triconfocal oval-cylinder coordinates	α, β, z	$\alpha \in \mathcal{D}_\alpha$	$\beta \in \mathcal{D}_\beta$	$z \in \mathbb{R}$
Triconfocal ovoid coordinates	$\tilde{\alpha}, \tilde{\beta}, \phi$	$\tilde{\alpha} \in \mathcal{D}_\alpha$	$\tilde{\beta} \in \mathcal{D}_\beta$	$\phi \in [0, 2\pi)$

[2] $\mathrm{Iso}(M, \mathcal{G}) := \{\phi : M \to M \mid \phi^* \mathcal{G} = \mathcal{G}\}$ endowed with the composition operation, where M is the manifold, $\mathcal{G}$ is the metric tensor and ϕ^* is the pushforward of ϕ.

projection of the metric tensor over the orthogonal hypersurfaces of w is taken as

$$(g_{ab}+w_a w_b)dx^a\otimes dx^b = f^2(\Lambda_u^2+\Upsilon_u^2)du\otimes du+k^2\left[\Upsilon^2(\Lambda_u^2+\Upsilon_u^2)dv\otimes dv+\Upsilon^2 d\Phi\otimes d\Phi\right]\,, \quad (54)$$

where f and k are functions of u and v, and $d\Phi = d\phi-(V/U)dt$. From here one can deduce the most general form of a metric tensor that satisfies the equation 54, which is

$$\begin{aligned}\mathcal{G} = g_{00}dt\otimes dt + g_{03}(dt\otimes d\phi + d\phi\otimes dt) + A^2 d\phi\otimes d\phi + f^2(\Lambda_u^2+\Upsilon_u^2)du\otimes du \\ + k^2\Upsilon^2(\Lambda_u^2+\Upsilon_u^2)dv\otimes dv\,. \end{aligned} \quad (55)$$

Over the 2-surfaces spanned by the vectors ∂_v and ∂_ϕ, the induced metric is conformal to that of an ovoid (see equation 49). To simplify the notation, the tilde emphasis in u and v has been omitted. Equation 54 shows that there is a part of the induced metric that is conformal to the metric over confocal ovoids, with k^2 being the conformal factor. The same procedure describe before can be implemented to derive a triconfocal ovoid metric ansatz $\mathcal{G}'$. For simplicity, this article focus only on this confocal case. Now it is convenient to write the static case of the metric $\mathcal{G}$ because it will be useful later. This corresponds to the metric

$$\mathcal{G}_{static} = g_{00}dt\otimes dt + k^2\Upsilon^2 d\phi\otimes d\phi + f^2(\Lambda_u^2+\Upsilon_u^2)du\otimes du + k^2\Upsilon^2(\Lambda_u^2+\Upsilon_u^2)dv\otimes dv\,. \quad (56)$$

It should be mention that by considering the equations 35 and 36 to write Λ_u and Υ_u in terms of $\Lambda = z$ and $\Upsilon = \sqrt{x^2+y^2}$ and by fixing $a = 0$ or $\tau = 0$, $\mathcal{G}_{static}$ in spherical coordinates is reduced to

$$\mathcal{G}_{spheric} = g_{00}dt\otimes dt + k^2 r^2\sin^2\theta\, d\phi\otimes d\phi + f^2 dr\otimes dr + k^2 r^2 d\theta\otimes d\theta\,. \quad (57)$$

which is a suitable metric ansatz to describe static spherically symmetric spacetimes. Hence, the *ovoid spacetimes* here described have the spherically symmetric spacetimes as a particular case. The condition $a = 0$ or $\tau = 0$ makes the u-constant surfaces in the ovoid coordinate system become concentric spheres to the origin with radius $r = u$ or $r = u/(1+a)$, respectively.

VIII. ANISOTROPIC FLUID SOLUTIONS TO THE EINSTEIN'S FIELD EQUATIONS

The relevance to define spacetimes with new symmetries is that it opens the possibility to obtain new exact solutions to the Einstein field equations. If it is a vacuum solution,

it can be interpreted as the solution for a black hole ore one part of it can be interpreted as the external solution for an object with ovoid symmetry. And, if it is a fluid solution, one would obtain a model for the gravitational field inside an object with ovoid symmetry. Both cases are very interesting and stimulating. Sadly, the Einstein field equations get too complicated. For this article, only a static anisotropic fluid solution is obtained.

Consider a static ovoid spacetime govern by the metric $\mathcal{G}_{static}$. A suitable stress-energy tensor for an static anisotropic fluid is

$$\mathcal{T} = T^{ab}\partial_a \otimes \partial_b = (\mu + P_1)w \otimes w + P_1\mathcal{G}^{-1}_{static} + (P_2 - P_1)e_2 \otimes e_2 + (P_3 - P_1)e_3 \otimes e_3\,, \quad (58)$$

where w is the four-velocity of the fluid, μ is its mass-energy density, P_i is its pressure in the direction of e_i with $i = 1, 2, 3$, $\{e_\mu\} = \{w, e_1, e_2, e_3\}$ is the orthonormal tetrad of the observer with four-velocity w, and $\mathcal{G}^{-1}_{static}$ is the inverse metric. Naturally, the quantities μ and P_i are the ones measured by the comoving observer with the fluid. In the case $P_2 = P_3 = P_1$ one recovers the stress-energy tensor of a perfect fluid in its standard form. In natural units the Einstein field equations are

$$R_{ab} = 8\pi\left(T_{ab} - \frac{1}{2}Tg_{ab}\right)\,, \quad (59)$$

where T is the trace of $\mathcal{T}$ and $\mathcal{R} = R_{ab}\,dx^a \otimes dx^b$ is the Ricci tensor.

A solution to this equations is given by

$$P_1 = \frac{1}{8\pi\Upsilon^2(\Lambda_u^2 + \Upsilon_u^2)^2}\left(\mathcal{P} - A_u^2\Upsilon^2(\Lambda_u^2 + \Upsilon_u^2)e^{-2B}\right)\,, \quad (60)$$

$$P_2 = \frac{1}{8\pi\Upsilon^2(\Lambda_u^2 + \Upsilon_u^2)^2}\left\{\mathcal{L} - \left[(A_u + B_u)\Upsilon_u - A_u^2\Upsilon\right](\Lambda_u^2 + \Upsilon_u^2)\Upsilon e^{-2B}\right\}\,, \quad (61)$$

$$P_3 = \frac{1}{8\pi\Upsilon^2(\Lambda_u^2 + \Upsilon_u^2)^3}\left\{\mathcal{Q} + \left[A_u^2\Upsilon^2(\Lambda_u^2 + \Upsilon_u^2) - \mathcal{L}\right](\Lambda_u^2 + \Upsilon_u^2)e^{-2B}\right\}\,, \quad (62)$$

$$\begin{aligned}\mu = \frac{1}{8\pi\Upsilon^2(\Lambda_u^2 + \Upsilon_u^2)^3}\big\{ &- \mathcal{Q} + \big[2\Upsilon^2 A_{uu} + (B_u + 5A_u)\Upsilon\Upsilon_u \\ &- A_u\Upsilon^2(2B_u + 3A_u) + \mathcal{P}\big](\Lambda_u^2 + \Upsilon_u^2)^2 e^{-2B}\big\}\,, \end{aligned} \quad (63)$$

$$g_{00} = e^{2A}\,, \quad (64)$$

$$f = e^{2B}\,, \quad (65)$$

$$k = e^{2(c_1 - A)}\,, \quad (66)$$

where A and B are arbitrary functions of u, c_1 is a constant, and $\mathcal{P}$, $\mathcal{Q}$ and $\mathcal{L}$ are given by

$$\mathcal{P} = \Upsilon_u \left(\Upsilon_{uv}\Lambda_u - \Lambda_{uv}\Upsilon_u\right)\left(e^{2(A-c_1)} - e^{-2B}\right), \tag{67}$$

$$\begin{aligned}\mathcal{Q} = &\left[(\Lambda_u^2 - \Upsilon_u^2)(\Upsilon_{uv}^2 + \Lambda_{uv}^2) + (\Lambda_u\Lambda_{uvv} + \Upsilon_u\Upsilon_{uvv})(\Lambda_u^2 + \Upsilon_u^2) - 4\Lambda_u\Upsilon_u\Lambda_{uv}\Upsilon_{uv}\right] \\ &\times \left(e^{2(A-c_1)} - e^{-2B}\right) + (A_u + B_u)(\Lambda_u^2 + \Upsilon_u^2)(\Lambda_u\Upsilon_{uv} - \Upsilon_u\Lambda_{uv})\Upsilon e^{-2B}, \end{aligned} \tag{68}$$

$$\mathcal{L} = \Lambda_u \left(\Lambda_{uv}\Lambda_u + \Upsilon_{uv}\Upsilon_u\right)\left(e^{-2B} - e^{2(A-c_1)}\right), \tag{69}$$

In order to check that this solutions satisfies the equation 59 it is convenient to use the relations shown in the section X D to simplify the expressions that involve derivatives of Λ and Υ. In this scenario, the weak and strong energy conditions can be satisfied if $\mu, P_i \geq 0$. In the limit when $x, y, z \gg \tau$ this inequalities take the approximate form

$$uA_u^2 - (A_u + B_u) \geq 0, \tag{70}$$

$$1 - e^{2(A+B-c_1)} - u^2A_u^2 \geq 0, \tag{71}$$

$$2A_u - uA_uB_u + uA_{uu} - uA_u^2 \geq 0. \tag{72}$$

These were obtained by expressing the derivatives of Λ and Υ in terms of themselves (see the appendix X D) and taking the high order terms in x, y and z. A solution to the exact or the approximate energy conditions is left for a future work.

IX. CONCLUSIONS

[1] A. Silva-Lora and R. Torres. Explicit cartesian oval as a superconic surface for stigmatic imaging optical systems with real or virtual source or image. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 476(2235):20190894, 2020.

[2] R. Descartes. *Discours de la méthode pour bien conduire sa raison et chercher la vérité dans les sciences, with three appendices: La Dioptrique (Les Météores, andLaGéométrie)*. Leyden, 1637.

[3] J. P. Southall. Aplanatic (or Cartesian) optical surfaces. *Journal of the Franklin Institute*, 193(5):609–626, 1922.

[4] A. Silva-Lora and R. Torres. Aplanatism in stigmatic optical systems. *Optics Letters*, 45(23):6390–6393, 2020.

[5] A. Silva-Lora and R. Torres. Rigorously aplanatic descartes ovoids. *J. Opt. Soc. Am. A*, 38(8):1160–1169, 2021.

[6] A. Silva-Lora and R. Torres. Achromatic stigmatism: achromatic cartesian ovoid. *J. Opt. Soc. Am. A*, 39(9):1524–1532, 2022.

[7] A. W. Greynolds. Superconic and subconic surface descriptions in optical design. In *International Optical Design Conference*. Optical Society of America, Optica Publishing Group, 2002.

[8] S. Cho. Explicit superconic curves. *J. Opt. Soc. Am. A*, 33(9):1822–1830, 2016.

[9] A. Silva-Lora and R. Torres. Superconical aplanatic ovoid singlet lenses. *J. Opt. Soc. Am. A*, 37(7):1155–1165, 2020.

[10] K. Schwarzschild. Untersuchungen zur geometrischen Optik II. *Astronomische Mittheilungen der Königlichen Sternwarte zu Göttingen*, 10:3–28, 1905.

[11] M. Chasles. *Aperçu historique sur l'origine et le développement des méthodes en géométrie*. Mém. couronnés, Acad. des sci. et belles lett. de Bruxelles. M. Hayez, Imprimeur de l'académie royale, 1837.

[12] B. Williamson. *An Elementary Treatise on the Differential Calculus: Containing the Theory of Plane Curves, with Numerous Examples*. Longmans, Green, and co., 1912. New impression.

[13] W. W. Johnson. Bipolar equations - cartesian ovals. *The Analyst*, 2(4):106–115, 1875.

[14] K. Schwarzschild. Über das gravitationsfeld einer kugel aus inkompressibler flüssigkeit nach der einsteinschen theorie. *Sitzungsberichte der königlich preußischen Akademie der Wissenschaften zu Berlin*, pages 424–434, 1916.

[15] R. C. Tolman. Static solutions of einstein's field equations for spheres of fluid. *Physical Review*, 55(4):364, 1939.

[16] H. D. Wahlquist. Interior solution for a finite rotating body of perfect fluid. *Physical Review*, 172(5):1291, 1968.

[17] F. J. Chinea and L. M. González-Romero. Interior gravitational field of a stationary, axially symmetric perfect fluid in irrotational motion. *Classical and Quantum Gravity*, 7(5):L99, 1990.

[18] A. Krasiński. Ellipsoidal space-times, sources for the kerr metric. *Annals of Physics*, 112(1):22–40, 1978.

[19] I. Racz. Note on stationary-axisymmetric vacuum spacetimes with inside ellipsoidal symmetry. *Classical and Quantum Gravity*, 9(8):L93, 1992.

[20] I. Rácz and M. Süveges. On the uniqueness of stationary axisymmetric ellipsoidal vacuum configurations. *Modern Physics Letters A*, 13(14):1101–1107, 1998.

[21] J. Zsigrai. Ellipsoidal shapes in general relativity: general definitions and an application. *Classical and Quantum Gravity*, 20(13):2855, 2003.

[22] C. W. Misner, K. S. Thorne, and J. A. Wheeler. *Gravitation*. Princeton University Press, 2017.

[23] H. Stephani, D. Kramer, M. MacCallum, C. Hoenselaers, and E. Herlt. *Exact Solutions of Einstein's Field Equations*. Cambridge Monographs on Mathematical Physics. Cambridge University Press, 2 edition, 2003.

[24] R. Ferréol. Cartesian Oval. [website]. https://mathcurve.com/courbes2d.gb/descartes/descartes.shtml (accessed: 02.07.2026).

[25] P. H. Moon and D. E. Spencer. *Field Theory Handbook: Including Coordinate Systems, Differential Equations, and Their Solutions*. John T. Zubal, 1988.

X. APPENDIX

A. Existence condition for Cartesian ovals

The existence conditions of ovals may be simplified by introducing the set $\mathcal{D}_u$ given by

$$\mathcal{D}_u(a;\tau) := \begin{cases} [\min(1,a)|\tau|, \infty) & \text{if } a > -1\,; \\ [-|\tau|, |\tau|] & \text{if } a = -1\,; \\ (-\infty, |\tau|] & \text{if } a < -1\,. \end{cases} \tag{73}$$

B. Orthogonality condition for coordinates q_1, q_2

Here, it is shown that any two coordinates q_1, q_2 are orthogonal if and only if their coordinate curves meet at right angles. To do so, consider the relations $q^i = q^i(x,y)$ and their gradient $\boldsymbol{\nabla} q^i$. Since $\boldsymbol{\nabla} q^i$ is orthogonal to the level curves of q^i then the coordinate curves intersect at right angles if

$$\boldsymbol{\nabla} q^1 \cdot \boldsymbol{\nabla} q^2 = q^1_x q^2_x + q^1_y q^2_y = 0\,, \tag{74}$$

where the subscripts indicate partial differentiation, $q^i_x := \partial q^i/\partial x$ for each $i = 1, 2$. By the chain rule one has that

$$\frac{\partial q^i}{\partial q^j} = q^i_x x_{,j} + q^i_y y_{,j} \,, \tag{75}$$

for each $i = 1, 2$ and $j = 1, 2$. For $i \neq j$, $\partial q^i/\partial q^j = 0$. Hence, in this case, one gets that $q^i_x = -q^i_y y_{,j}/x_{,j}$, which implies that $\boldsymbol{\nabla} q^1 \cdot \boldsymbol{\nabla} q^2 = q^1_y q^2_y (y_{,1} y_{,2} + x_{,1} x_{,2})/x_{,1} x_{,2}$, so that equation 8 implies equation 74, and vice versa.

C. The s-equation as an algebraic equation in some power of s

D. Characterization of the second and third derivatives of the metric functions Λ and Υ